THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. How many non-square numbers lie between 9^2 and 10^2? 1

(a) 9 **(b)** 18 **(c)** 19 **(d)** 17

2. The value of $\dfrac{1}{2^{-3}}$ is: 1

(a) $\frac{1}{8}$ **(b)** 6 **(c)** -8 **(d)** 8

3. Which of the following is NOT a correct Roman numeral? 1

(a) XL **(b)** IX **(c)** IC **(d)** IV

4. The angles of a quadrilateral are in the ratio 1:2:3:4. The largest angle is: 1

(a) 156° **(b)** 144° **(c)** 132° **(d)** 108°

5. Which of the following numbers is divisible by 11? 1

(a) 3464 **(b)** 3465 **(c)** 3466 **(d)** 3467

6. $(x+3)^2$ equals: 1

(a) x^2+9 **(b)** x^2+3x+9 **(c)** x^2+6x+9 **(d)** x^2+6x

7. The ratio of 2 m 40 cm to 60 cm is: 1

(a) 5 : 1 **(b)** 4 : 1 **(c)** 4 : 2 **(d)** 1 : 4

8. A value changes from 80 to 100. The percentage increase is: 1

(a) 28% **(b)** 25% **(c)** 22% **(d)** 50%

9. The legs of a right triangle are 5 cm and 7 cm. Its hypotenuse (in cm) is: 1

(a) $5\sqrt{3}$ **(b)** $2\sqrt{3}$ **(c)** $\sqrt{74}$ **(d)** 12

10. If A:B = 2:3 and B:C = 4:5, then A:B:C is: 1

(a) 8:15:8 **(b)** 2:3:5 **(c)** 2:7:5 **(d)** 8:12:15

11. A polyhedron has 12 vertices and 18 edges. The number of faces is: 1

(a) 8 **(b)** 6 **(c)** 32 **(d)** 28

12. The median of 7, 18, 50, 6, 14, 44, 22, 5 is: 1

(a) 16 **(b)** 20.75 **(c)** 18 **(d)** 14

13. Think of a number n. Add 3, multiply by 5, then subtract 15. The result is: 1

(a) $n+3$ **(b)** $5n+3$ **(c)** $3n$ **(d)** $5n$

14. The area of a rectangle is 360 cm² and its length is 24 cm. Its breadth (in cm) is: 1

(a) 15 **(b)** 336 **(c)** 17 **(d)** 180

15. The value of $15^2 - 14^2$ is: 1

(a) 28 **(b)** 29 **(c)** 15 **(d)** 30

16. The value of $(-3)^4$ is: 1

(a) -81 **(b)** 12 **(c)** 85 **(d)** 81

17. The decimal value of 243_5 is: 1

(a) 78 **(b)** 73 **(c)** 75 **(d)** 71

18. In a parallelogram ABCD, $\angle A = (2x + 40)^\circ$ and $\angle C = (3x + 10)^\circ$. The value of x is: 1

(a) 31 **(b)** 35 **(c)** 30 **(d)** 29

19. Which of the following numbers is divisible by 6? 1

(a) 4519 **(b)** 4518 **(c)** 4517 **(d)** 4520

20. $(x - 2)^2$ equals: 1

(a) $x^2 - 4$ **(b)** $x^2 - 4x + 4$ **(c)** $x^2 - 2x + 4$ **(d)** $x^2 + 4x + 4$

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Express 100 as the sum of consecutive odd numbers starting from 1. 2

22. Express $(2^3)^4 \times 2^{-5}$ as a single power of 2. 2

23. Convert 314_5 to a decimal number. 2

24. The angles of a quadrilateral are in the ratio 1:3:5:3. Find the smallest angle. 2

25. Use the divisibility rule for 11 to test 6578. 2

26. Expand $(x + 8)^2$ using an identity. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Show that $T_5 + T_6$ is a perfect square. 3

28. Express each as a single power of 2: 3

(i) $2^5 \times 2^{-2}$

(ii) $(2^3)^{-2}$

(iii) $2^4 \div 2^{-3}$.

29. Convert 137 to base 5 and check your answer by converting back to decimal. 3

30. The angles of a quadrilateral are in the ratio 1:2:3:4. Find all the angles. 3

31. Check whether 35640 is divisible by 8 and by 11. 3

OR

Expand $(2x + 3)^2$. 3

32. If A : B = 2 : 3 and B : C = 6 : 5, find A : B : C. 3

OR

The marked price of a laptop is ₹ 45000. GST at 18% is added. Find the GST amount and the final bill. 3

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SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Verify that 1729 can be written as the sum of two cubes in two different ways: $1^3 + 12^3$ and $9^3 + 10^3$. **5**

34. Find x if **5**

(i) $2^x \times 2^3 = 2^{11}$

(ii) $(2^x)^4 = 2^{20}$

(iii) $2^{x-2} = \dfrac{1}{2^3}$.

35. Add the binary numbers 1101_2 and 111_2. Verify your answer by converting to decimal. **5**

36. (i) Show that the sum of the angles of a quadrilateral is $360°$. **5**

(ii) Three angles of a quadrilateral are $78°$, $92°$ and $104°$. Find the fourth angle.

OR

Replace the stars so that the number 4*3* is divisible by both 5 and 9. Find all possibilities. **5**

37. A rectangle has length $(x + 3)$ cm and breadth $(x + 5)$ cm. Write an expression for its area and find the area when $x = 4$. **5**

OR

Divide ₹ 1200 among A, B and C in the ratio 1 : 3 : 4. If C gives ₹ 100 to A, find the new ratio of their amounts. **5**

38. Find the compound interest on ₹ 8000 for 3 years at 10% per annum compounded annually. By how much does it exceed the simple interest for the same period? **5**

OR

The diagonals of a rhombus are 40 cm and 42 cm. Find its side, perimeter and area. **5**

END OF QUESTION PAPER — **Total: 80 Marks**

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